

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.2s gimbal=0.0363393637s cost=1.4230251357 lag_gradient={'rotor':
-0.9412990583541921, 'gimbal': -2.3927212912944196e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.3877787807867628e-17,
'analytic_l2_norm': 8.008531755445352, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 8.008531755445352, 'relative_error': 1.0}, {'direction':
'rotor_delay_seconds', 'scheme': 'forward', 'step': 1.3877787807814457e-17,
'analytic_l2_norm': 8.008534533684392, 'finite_difference_l2_norm': 0.0,
'difference_l2_norm': 8.008534533684392, 'relative_error': 1.0}, {'direction':
'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 16.8658854912103, 'finite_difference_l2_norm': 16.865885383350914,
'difference_l2_norm': 7.121923147717956e-07, 'relative_error': 4.222679652028685e-08}]
smooth width=2: rotor=0.2s gimbal=0.0362132886s cost=1.42434107082 lag_gradient={'rotor':
-0.9404029761187671, 'gimbal': 2.094436124033816e-06} FD=[{'direction': 'negative_gradient',
'scheme': 'forward', 'step': 1.387778780785048e-17, 'analytic_l2_norm': 8.87141132533677,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 8.87141132533677, 'relative_error':
1.0}, {'direction': 'rotor_delay_seconds', 'scheme': 'forward', 'step':
1.3877787807814457e-17, 'analytic_l2_norm': 8.871409422212087, 'finite_difference_l2_norm':
0.0, 'difference_l2_norm': 8.871409422212087, 'relative_error': 1.0}, {'direction':
'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 18.698381366551164, 'finite_difference_l2_norm': 18.698380287322877,
'difference_l2_norm': 4.9544839386444546e-06, 'relative_error': 2.649686003039464e-07}]
smooth width=1: rotor=0.2s gimbal=0.0360214911s cost=1.42482254224 lag_gradient={'rotor':
-0.917246377556522, 'gimbal': 0.012207461040635526} FD=[{'direction': 'negative_gradient',
'scheme': 'forward', 'step': 1.3879016833349834e-17, 'analytic_l2_norm': 11.126164649238538,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 11.126164649238538,
'relative_error': 1.0}, {'direction': 'rotor_delay_seconds', 'scheme': 'forward', 'step':
1.3877787807814457e-17, 'analytic_l2_norm': 11.114743282612935, 'finite_difference_l2_norm':
0.0, 'difference_l2_norm': 11.114743282612935, 'relative_error': 1.0}, {'direction':
'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 24.965198596794203, 'finite_difference_l2_norm': 24.926697934252708,
'difference_l2_norm': 0.9522899942078953, 'relative_error': 0.038144699330778783}]
smooth width=0.5: rotor=0.2s gimbal=0.0382506084s cost=1.42476671875 lag_gradient={'rotor':
-0.8764106958392942, 'gimbal': -4.440797272825536e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'forward', 'step': 1.387778780799288e-17, 'analytic_l2_norm':
14.846386690678184, 'finite_difference_l2_norm': 0.0, 'difference_l2_norm':
14.846386690678184, 'relative_error': 1.0}, {'direction': 'rotor_delay_seconds', 'scheme':
'forward', 'step': 1.3877787807814457e-17, 'analytic_l2_norm': 14.846384126769957,
'finite_difference_l2_norm': 0.0, 'difference_l2_norm': 14.846384126769957,
'relative_error': 1.0}, {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step':
6.0554544523933395e-06, 'analytic_l2_norm': 21.45630069584428, 'finite_difference_l2_norm':
21.456257979847678, 'difference_l2_norm': 0.00014732715825392678, 'relative_error':
6.866382063822471e-06}]

strict profile iterations: 1

iteration 1: best=(0.2, 0.038250608431) cost=1.42487427189 rotor_range=(0.195002269745, 0.2)
gimbal_range=(0.033251924502, 0.043249292361) expansions=[]

selected rotor lag 0.2 s (40.0182 publish periods)

selected gimbal lag 0.038250608431 s (7.65214 publish periods)
selected objective cost 1.42487422806